

Sidecar Build Kit

The spike to run this week, the one Android restriction that can sink the idea, and thirty hours broken into blocks with a cut line.

Spike window · 16–21 Sep Clock starts · Sat 26, 11:00

Two scored reviews · Sat eve, Sun morn

Read this before you write a line of code

THE ONE THING THAT CAN KILL SIDECAR

Since Android 10, an app cannot read the clipboard unless it has focus. The picture of a phone quietly watching the clipboard in the background does not work, and no permission unlocks it. Discovering this at hour six on Saturday would be fatal, so design around it now.

Three legal paths exist, and the rules of this hackathon make the first one perfectly acceptable:

PATH	HOW IT WORKS	VERDICT
Foreground app	Sidecar is open on the phone. It reads the clipboard on focus and on a tap, processes, writes the answer back. Office Kit syncs that clipboard to the laptop.	Use this. The rules require you to run and pitch on the phone anyway, so the app being in the foreground is the intended posture, not a compromise.
Share sheet	Register an <code>ACTION_SEND</code> and <code>ACTION_PROCESS_TEXT</code> intent filter. Selecting text anywhere on the phone offers "Sidecar" in the menu.	Add on Sunday if time allows. Costs little, looks polished, and shows Android fluency to the jury.
Watched folder	Office Kit file transfer drops a file; the app polls a directory chosen through the Storage Access Framework.	Build this too — it is the fallback if clipboard sync turns out to be one-directional on the loaner.

REFRAME FOR THE PITCH

Do not apologise for the foreground tap. Say it deliberately: *"Nothing reads your clipboard behind your back — you decide what crosses to the model, one tap at a time."* The OS restriction becomes a privacy feature, which is what the whole product is about.

The spike, Wednesday to Monday

SIX EVENINGS · EACH ONE ENDS IN A YES OR A NO

WED 16

~1 HOUR

Run a local model on any Android phone

Install Google AI Edge Gallery or PocketPal on whatever Android device is to hand, download a small model, and talk to it with wifi off. No code.

This is the cheapest hour of the week: it makes "Experimented with local LLMs" a true answer on the submission form, and it calibrates your sense of how fast a 1B model actually replies on a phone.

SHIPS: AN HONEST PROFICIENCY ANSWER

THU 17

~3 HOURS

A dev build that loads a model

ExecuTorch ships native code, so Expo Go will not run it — you need a development build. Peer dependencies are explicit in the package, so install them together:

```
# new app + dev client
npx create-expo-app@latest sidecar
cd sidecar
npx expo install expo-dev-client

# the library and its four peer deps
npx expo install react-native-executorch \
  react-native-worklets \
  react-native-blob-util \
  @kesha-antonov/react-native-background-downloader

# native project + run on a real device (not an emulator)
npx expo prebuild --platform android
npx expo run:android --device
```

Then the smallest possible screen: the `useLLM` hook pointed at one of the bundled model constants, a text input, and the streamed `response` on screen. Watch `downloadProgress` the first time — the model is a large download and you do not want that happening on venue wifi.

GATE: A MODEL ANSWERS ONE PROMPT ON-DEVICE

FRI 18
~3 HOURS

Prove the bridge, both halves

Clipboard: read on focus, write back, with `@react-native-clipboard/clipboard`. Test the loop by hand — copy on a PC, bring the app to the front, confirm it sees the text, write a transformed string back, confirm the PC can paste it.

Folder: an SAF directory picker plus a poll for new files. `react-native-blob-util` is already installed as a peer dependency and reads the picked file.

GATE: TEXT MAKES A ROUND TRIP

FRI 18, LATE
DECISION

Go / no-go

If Thursday and Friday both landed, Sidecar is de-risked and the rest of the weekend is content. If the model would not load on a real device, switch the submission to **Paper Desk** — camera plus ML Kit OCR plus the same local model, no bridge dependency at all. Same deck skeleton, two slides rewritten.

Do not carry an unproven core into Sunday's deck build.

SAT 19
~2 HOURS

Record sixty seconds

Phone screen recording. Airplane mode visibly on in the status bar. Load the model, paste a paragraph of a contract, show the summary appear. No face, no slides, no music, no editing beyond a trim. Upload unlisted and paste the link into the video field.

SHIPS: THE FIELD ALMOST NOBODY FILLS

SUN 20
~2 HOURS

Finish the deck, fill the form

The deck is already built — check the architecture slide still matches what Friday proved, and swap in a real screenshot from the spike if you have one. Paste the description, prior-builds and standout answers. Tick the originality box and mention the spike honestly as a throwaway prototype.

MON 21
~20 MIN

Submit a day early

Only the team leader can submit, and the platform will be under load on the 22nd. Submit, screenshot the confirmation, then stop touching it.

ORIGINAL-WORK RULE

Code must be written inside the event window; a pre-built app is disqualification. Keep this week's work a throwaway spike in a separate repository, disclose it on the originality checkbox, and start a clean repo at 11:00 on Saturday. What carries over legitimately is *knowledge* — which commands work, which model size is fast enough, where the clipboard bites.

Thirty hours

SAT 26 10:00 – SUN 27 17:00 · BLOCKS, GATES AND A CUT
LINE

Two scored evaluation rounds sit inside the window, so something has to be demoable by Saturday evening and something better by Sunday morning. Build in that order, not in architectural order.

SAT 08:00

CHECK-IN

Collect the loaner, pair Office Kit

Ask two questions at the 10:00 teach-in, before anyone else queues: **does clipboard sync run both ways**, and **where does file transfer land on the phone**. Both answers change what you build first.

SAT 11:00

HOURS 0–2

Model on the loaner, nothing else

Fresh repo, dev build, model downloaded and answering. Download the model over your own hotspot if venue wifi is slow — this is the only step where a network stall can cost you the weekend.

GATE: INFERENCE ON THE LOANER BY 13:00

SAT 13:00

HOURS 2–6

The clipboard round trip

Read on focus, one prompt template, write back. A single screen: what came in, what went out, a token counter. Ugly is fine. This alone is a complete demo.

GATE: DEMOABLE PRODUCT EXISTS BY 17:00

SAT 17:00

HOURS 6–8

Make it presentable, then present it

Two hours of interface before the first review, not two minutes. The jury scores end-product quality at 30% and they are looking at the screen. Then the 19:00 evaluation: lead with airplane mode.

SAT 20:00

HOURS 9–13

Camera and OCR

Point at a printed page, ML Kit extracts text, same pipeline answers. Built during the Red Light window, so build it phone-only on purpose — it is the part that needs no laptop.

SHIPS: THE CREATIVE-PHONE-USE COUNTER

SUN 00:00

HOURS 13–17

Voice in, voice out

Whisper for speech, the library's text-to-speech for answers. Hands-free is what makes the thing feel like a co-processor rather than a text box.

SUN 04:00

HOURS 17–20

Sleep, then the local index

Take three hours. A tired solo builder loses more than the feature is worth. Then embeddings over everything ingested, so it can answer across two documents — this is what turns four demos into one product.

GATE: FULL LOOP BY 09:00 REVIEW

SUN 09:00

HOURS 22–24

Second evaluation, then the NPU port

Move the hot model to the QNN execution provider and measure first-token latency before and after. Keep the working CPU path behind a flag; port on a branch. Two numbers on a slide is the whole technical-depth argument.

CUT LINE: IF IT IS NOT WORKING BY 12:30, REVERT AND STOP

SUN 12:30

HOURS 25–28

Freeze and rehearse

No new features. Fix only what breaks the demo path. Rehearse the three minutes twice, out loud, on the phone, with airplane mode on. Time it. Submit the repo before the cutoff — repos lock before the Top 10 pitches.

SUN 15:00

PITCH

Top 10

Airplane mode already on when you walk up. Do not spend the first minute explaining architecture — show the round trip, then explain why it could only work this way.

THE CUT LINE

If you are behind, drop in this order: the local index, then voice output, then the NPU port. Never drop the camera path — it is a scored counter and it is thirty minutes of work. Never drop interface polish; it is the single heaviest line on the rubric.

Three minutes

PITCH SCRIPT · REHEARSE IT TWICE

0:00

"This phone has been in airplane mode since before I walked up here. It stays that way for the next three minutes."

Hold the phone up, status bar visible. Do not explain the product yet.

0:20

"Every AI tool on my work laptop is a browser tab that ships my client's contract to someone else's servers. So the documents that matter most are the ones I get no help with."

One sentence of problem. Resist the temptation to add a second.

0:40

"Watch the laptop."

Copy an indemnity clause. Count out loud if it takes two seconds. Paste the answer back into the document. This is the moment the room decides.

1:20

"This annexure is printed. Nothing to copy."

Camera on the page, OCR, spoken answer. Let the phone talk — silence from the presenter while the device speaks is the strongest beat in the pitch.

1:50

"Now ask it something that needs both documents."

The local index answers. Point at the byte counter, still zero.

2:20 "Llama 3.2 on the Hexagon NPU through QNN. First token dropped from X to Y milliseconds when we ported it this morning. Office Kit carries the text; nothing else leaves the two devices."

Real numbers, measured. This is the technical-depth slot — thirty seconds, no more.

2:50 "The phone in your pocket is the only device on your desk that can be trusted with the documents you cannot upload. That is the product."

Stop talking. Leave the byte counter on screen for questions.

Carry list

VENUE

- **Model files on a USB drive and on the laptop.** Venue wifi at a 100-team hackathon will not give you a multi-gigabyte download twice.
 - **A printed page with real contract text on it** for the camera demo. Not lorem ipsum — the OCR needs to read something a judge can verify.
 - **Your own hotspot** for the first hour only.
 - **USB-C cable and a powered hub.** The loaner runs hot and flat under sustained inference; thermals are a real constraint and worth naming in the pitch.
 - **A written fallback plan** on one page, so a 03:00 decision is made by the version of you that is awake now.
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References

- [react-native-executorch documentation](#) — useLLM API, model constants, supported tasks
- npm registry — `react-native-executorch@0.10.2` peer dependencies, verified 16 Sep 2026
- iQOO Hackathon 2026 — [Guide & Rules](#) — evaluation rounds, Red Light / Green Light, repo lock, original-work rule